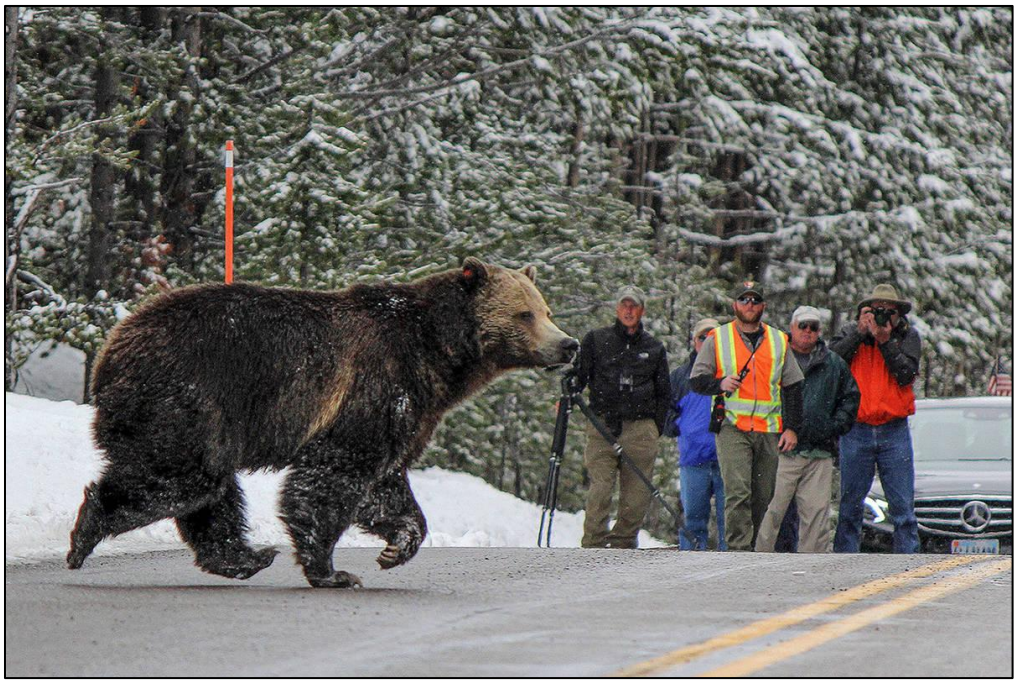
ENVRSUST 9014 Practice R Exam

Human and wildlife conflict

Description: National parks provide both protection for biodiversity and recreational ecosystem services for society. However, this can sometimes lead to conflict between humans and wildlife. As a park manager, you want to identify major issues with wildlife and human conflict using reporting data. The datasets we will be using contain the incident type, incident date, the species involved, and whether or not a deterrent was used.



Datasets to work with (in data folder):

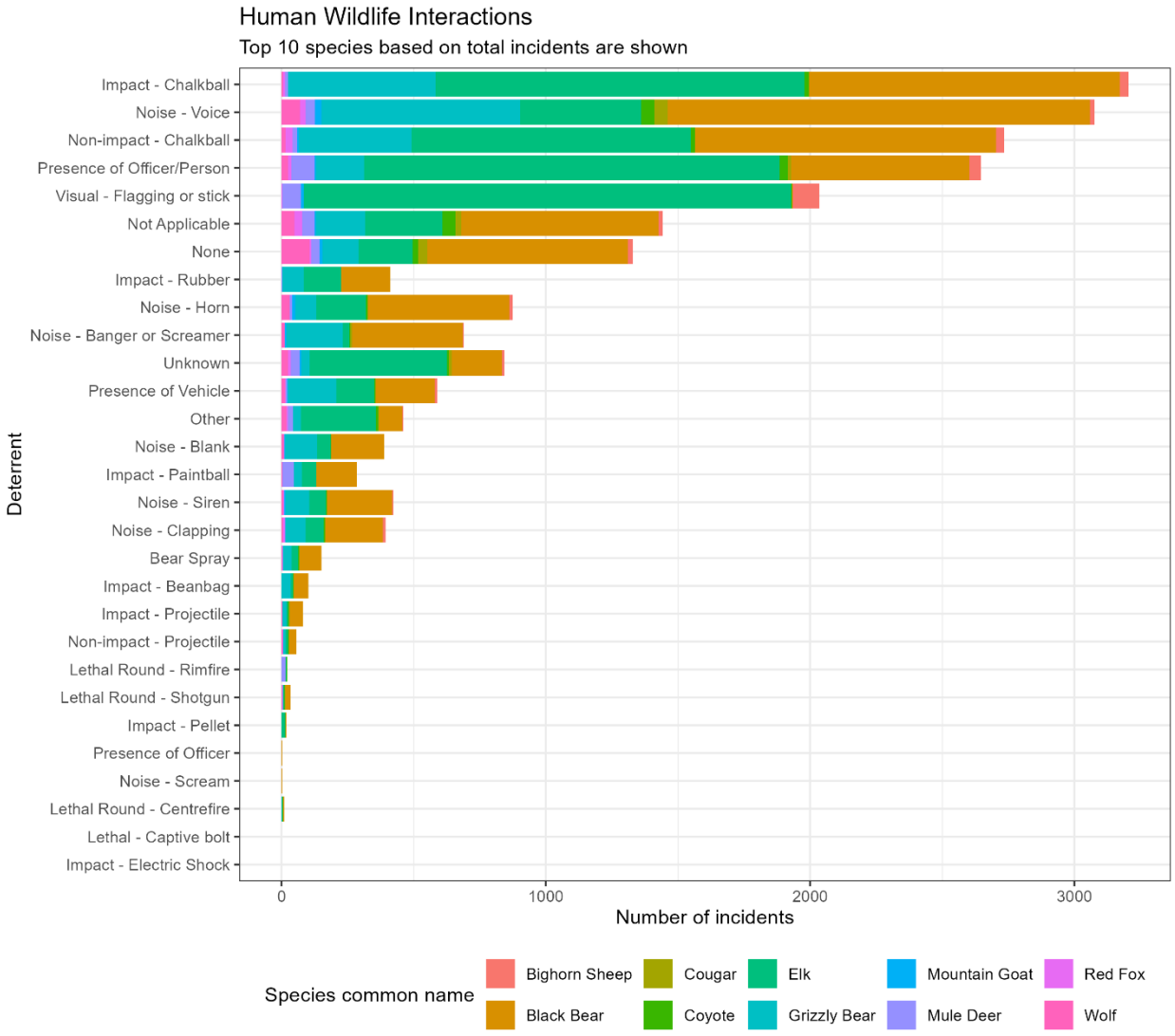
Dataset	Description
pca-human-wildlife-coexistence-animals-involved-detailed-records-2010-2023.csv	Each row in the dataset is a recorded incident with a different species
pca-human-wildlife-coexistence-incidents-detailed-records-2010-2023.csv	This dataset has staff hours across years according to incident type

1. In order to better plan and prepare for interventions, we need to examine how incidents are occurring over time. This might allow us to put more resources in place to respond, or limit, negative interactions. Use the pca-human-wildlife-coexistence-animals-involved-detailed-records-2010-2023.csv to recreate the following figure. Species can be found in the Species Common Name column. Remove species listings that are classified as "Unknown". Note, there are some species that have "unknown" in front of their common name (e.g., "unknown duck", "unknown mollusk", etc), please keep those in the dataset.

   Include code in your script to save your figure as a PNG.



2. Occasionally, a deterrent may have to be used when there is a human wildlife interaction to protect humans. Filter the pca-human-wildlife-conflict-animals-involved-detailed-records-data dataset for only 'Human Wildlife Interaction' (found in the Incident Type column). Once you have determined the top five species, go back and filter the deterrent_records dataset to only include those top 5 species and then recreate the figure below.



3. We want to learn more about how the number of animals involved in different incident types has changed over time. From the pca-human-wildlife-coexistence-animals-involved-detailed-records-2010-2023.csv, create a new dataset that only has the following columns:
   - Year
   - Incident Type
   - Sum of number of animals

   Recreate the following figure which explores how *the mean* number of animals per incident type has changed from 2013 to 2023 (hint: use back-ticks around the year headers when doing your calculations, `2013`).

